TraceBack — Requirements to Jira to GitHub to Testing Traceability Matrix

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All 22 functional requirement groups delivered in TraceBack and all 8 non-functional requirements trace from a requirement to its Jira story, its pull request, the code that implements it and the tests that check it, and every one passes on the final code. Deliverable 3 changes are shown as commits on the `d3/integration` branch.

TraceBack is the IIT Kanpur Lost & Found portal. Requirement IDs come from Deliverable 1 and the Deliverable 2 report (section 11). Jira keys are from the SCRUM project; the Jira site is currently suspended (free-plan member limit), so the keys are taken from the branch names and commit messages, which carry them. Pull requests are in github.com/cs455project/campus-lost-and-found.

How to read a row: automated tests are pytest test functions in `backend/tests/` (459 in total); SYS, ADV, CON and FR are the system, adversarial, concurrency and failure scenarios in the Test Reports, Concurrency and Failure and Recovery tabs.

Functional requirements

Requirement	Jira story	GitHub	Implementation	Automated tests (examples)	System and D3 tests	Result
FR-AUTH-01–04: sign in with an IITK email and a one-time code	SCRUM-10	PR #11; `c2b51ad`, `9fd794f`	`services/auth.py`, `routers/auth.py`	`test_full_sign_in_creates_a_student_account`, `test_expired_code_is_refused`, `test_not_signed_in_gets_401`, `test_a_code_that_could_not_be_sent_does_not_make_the_student_wait`	SYS-02, ADV-01–05, ADV-23, CON-01–03, FR-08	Pass
FR-PROF-01–03: onboarding and profile	SCRUM-35	PR #12	`services/users.py`	`test_phone_hall_room_editable_but_name_and_roll_locked`, `test_mobile_number_is_required_for_everyone`	SYS-02, ADV-08	Pass
FR-SRCH-01: browse lost and found listings	SCRUM-39	PR #13	`routers/items.py`, `ItemBrowser`	`test_found_item_hidden_until_received`, `test_pagination`	SYS-06, SYS-13	Pass
FR-REP-01, 03–08: report a lost item	SCRUM-16	PR #22; `9fd794f`	`services/items.py`	`test_report_lost_item`, `test_report_rejects_bad_values`, `test_a_new_report_is_saved_with_its_ai_job_in_one_commit`	SYS-03, ADV-10, FR-04	Pass
FR-REP-02–08: report a found item with photos	SCRUM-36	PR #23; `423b3bf`, `a541035`	`services/items.py`, `services/storage.py`	`test_report_found_item_waits_for_dropoff`, `test_fake_image_is_rejected`, `test_jpeg_loses_gps_and_xmp_but_keeps_its_picture_and_rotation`, `test_photo_with_too_many_pixels_is_refused`	SYS-04, ADV-15–18, CON-10	Pass
FR-REP-09: edit my report	SCRUM-38	PR #24	`update_report`	`test_only_the_reporter_can_edit`, `test_no_edits_after_closing_or_handing_in`	ADV-09	Pass

Requirement	Jira story	GitHub	Implementation	Automated tests (examples)	System and D3 tests	Result
FR-REP-10–12: close or withdraw my report	SCRUM-37	PR #24	`close_lost`, `cancel_found`	`test_close_lost_report`, `test_finder_can_cancel_before_handing_in`	ADV-09	Pass
FR-SRCH-02–06: search and filter	SCRUM-40	PR #25	`build_search_query`, `ItemBrowser`	`test_search_matches_word_forms`, `test_category_filter_and_counts`	SYS-06, ADV-12, ADV-13	Pass
FR-CUS-01–02: hand a found item to an SIS guard	SCRUM-17	PR #17	drop-off office in `services/items.py`	`test_finder_can_change_the_office_before_handing_in`, `test_dropoff_reminder_after_24_hours_once`	SYS-04, SYS-05	Pass
FR-CUS-03–05: guard receives and holds the item	SCRUM-45	PR #18; `c2b51ad`	`services/custody.py`	`test_receive_records_who_where_and_when`, `test_receive_only_once_and_only_open_offices`	SYS-05, CON-04, CON-11	Pass
FR-CLM-01–05: claim an item with private details	SCRUM-41	PR #26; `c2b51ad`, `a541035`	`services/claims.py`	`test_submit_claim_marks_item_claim_pending`, `test_proof_images_are_private`, `test_claim_proofs_are_cleaned_too`	SYS-07, ADV-09, ADV-11, CON-05	Pass
FR-CON-01–04: one approved claim per item	SCRUM-44	PR #26; `c2b51ad`	row locks and partial unique index	`test_database_allows_only_one_approved_claim`, `test_only_one_open_claim_per_person`	CON-06, CON-07	Pass
FR-CLM-07, 08: guard reviews and verifies claims	SCRUM-46	PR #27; `c2b51ad`	staff claim endpoints	`test_students_cannot_review`, `test_cannot_skip_steps`	SYS-08, ADV-06, CON-07	Pass
FR-CUS-07: final hand-over and Returned page	SCRUM-47	PR #28; `c2b51ad`	`hand_over`, `services/returned.py`	`test_hand_over_requires_id_check`, `test_hand_over_returns_item_and_closes_other_claims`	SYS-09, CON-08, CON-09	Pass
FR-AI-01–07: ranked AI match candidates	SCRUM-49	PR #14; `9fd794f`	`services/ai/semantic.py`, `queue.py`	`test_lost_report_gets_the_matching_found_item`, `test_only_the_top_k_are_kept`, `test_an_interrupted_job_is_run_again`	SYS-06, CON-13, FR-05, FR-06	Pass
FR-AI-08, 09: matching keeps working when the AI fails	SCRUM-50	PR #15; `a9aba2b`	`services/ai/agent.py`, `llm.py`	`test_provider_failure_falls_back_to_rule_based_matching`, `test_agent_cannot_save_ids_it_was_not_shown`, `test_when_the_agent_is_asked_follows_the_settings`	FR-07	Pass
FR-AI-11–18: AI never decides; ID-card owner alerts	SCRUM-52	PR #16; `b2b4cf1`	guardrails, `services/ai/idcard.py`	`test_the_agent_never_sees_the_identifiers`, `test_report_text_never_includes_personal_details`	ADV-09	Pass
FR-ADM-01, 03, 04: admin adds and manages staff	SCRUM-53	PR #19	`services/admin.py`	`test_admin_adds_another_admin_who_can_manage_staff`, `test_admin_can_add_a_gmail_guard_who_can_then_sign_in`	SYS-10, ADV-07	Pass
FR-ADM-02, 05–08: roles and their limits	SCRUM-54	PR #19	`services/admin.py`, `core/deps.py`	`test_guards_cannot_manage_staff`, `test_admin_cannot_demote_or_remove_themselves`	SYS-11, ADV-06, ADV-07	Pass
FR-LIFE-01–03: drop-off, collection and review reminders	SCRUM-19	PR #20; `a9aba2b`	`services/jobs.py`, `worker.py`	`test_dropoff_reminder_after_24_hours_once`, `test_collect_reminder_after_3_days_once`, `test_approval_email_never_shows_placeholder_text`	SYS-12, CON-12	Pass

Requirement	Jira story	GitHub	Implementation	Automated tests (examples)	System and D3 tests	Result
FR-LIFE-04, 05: 60-day retention and proof clean-up	SCRUM-55	PR #20	`services/jobs.py`	`test_returned_items_deleted_after_60_days`, `test_one_failing_job_does_not_stop_the_others`	SYS-12	Pass
FR-AUD-01–03: audit logs and operator tools	SCRUM-57	PR #21; `a9aba2b`	`email_logs`, `agent_runs`, `cli.py`	`test_every_attempt_is_logged_without_the_sign_in_code`, `test_a_temporary_failure_is_retried_and_the_email_goes_out`	FR-08, FR-09	Pass

Non-functional requirements

Requirement	Source	GitHub	Implementation	Automated tests (examples)	System and D3 tests	Result
NFR-SEC: only the right people see and change data; injection, XSS and malicious uploads are refused	D1; D2 plan	PRs #11, #19; `0e64287`, `a541035`	role dependencies, input schemas, `storage.py`, `core/headers.py`	`test_every_api_response_carries_the_security_headers`, `test_api_docs_can_be_switched_off`	ADV-01–25 (23 attacks stopped, 2 audits clean)	Pass
NFR-PRIV: no personal data to AI services; shared photos carry no location	D2 (NFR-PRIV)	PR #16; `a541035`, `b2b4cf1`	agent guardrails, `storage.strip_metadata`	`test_the_agent_never_sees_the_identifiers`, `test_uploaded_item_photos_reach_other_students_without_gps`	ADV-17	Pass
NFR-PERF: 100 daily users and 250+ transactions a day; pages under 3 s, API p95 under 500 ms	D2 plan for D3	`423b3bf`	`core/capacity.py`, thumbnails, fewer queries	`test_requests_over_the_limit_wait_then_get_server_busy`, `test_signed_in_request_uses_one_query_for_session_and_user`	SYS-15; load tests (ramp, campus, spike); CON-14	Pass
NFR-CONC: parallel actions never break the rules	D2 (FR-CON); D2 plan	PR #26; `c2b51ad`	row locks, advisory lock, `SKIP LOCKED`	`test_database_allows_only_one_approved_claim`	CON-01–14, 40 trials each	Pass
NFR-REL: recovers from database, process and outside-service failures	D2 plan for D3	`9fd794f`	503 handler, job re-run, restart policy	`test_an_unreachable_database_gives_a_clear_503_not_a_500`, `test_an_interrupted_job_is_run_again`	FR-01–10	Pass
NFR-AI-1: AI calls within the daily limit; clear matches use no AI	D2 (11.3)	PR #15	`AGENT_MATCHING=ambiguous`, `AI_DAILY_CALL_LIMIT`	`test_daily_limit_stops_ai_calls`, `test_no_photo_description_once_the_daily_ai_limit_is_reached`	FR-07	Pass
NFR-AI-2: pages never fail when the AI fails	D2 (11.3)	PR #15	rule-based fallback	`test_provider_failure_falls_back_to_rule_based_matching`	FR-07	Pass
NFR-DEP: runs on the AWS free plan, deployed with one command	D2 plan for D3	`c37f6d3`	`deploy/` (Docker Compose, Caddy, `upload.ps1`)	– (configuration)	Deployment checks; FR-10	Pass

Requirements not delivered

Requirement	Jira	Decision	Reason
FR-DIR-01–03: direct hand-over between finder and owner	SCRUM-42, 43	Dropped by design (D2, RC-05)	Every hand-over goes through an SIS guard, which is safer for valuables
FR-CHAT-01–03: chat between finder and claimant	SCRUM-56	Dropped (D2, RC-16)	Chat would leak the details used to verify ownership and invite social engineering
FR-SCORE-01–04: verification score for guards	SCRUM-51	Descoped in Deliverable 3	The guard's review with the claimant's private details and an ID check covers verification; the score was advisory only
FR-APT-01–05: hand-over time slots	SCRUM-18, 48	Descoped in Deliverable 3	Office hours, the 7-day collection window and "request an office visit" cover collection

Coverage of the chain

- Every delivered functional requirement has at least one automated test and one system-level or Deliverable 3 test; the deployment requirement is checked by the deployment checklist and the power-cut test (FR-10).
- Every story pull request (#11–#28) appears in the matrix, and every Deliverable 3 commit is linked to the requirements it changed.
- The 459 automated tests, 15 system scenarios, 23 attacks, 14 concurrency checks and 10 failure scenarios all pass on the final code (Test Reports, Concurrency, Failure and Recovery tabs).